

Flatness-Based and Passivity-Based Control with Adaptive Neural Compensation of a Direct-Drive PMSG Wind Energy Conversion System

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Abstract—This paper proposes a control strategy for a grid-connected 2 MW direct-drive permanent magnet synchronous generator (PMSG) wind turbine with a back-to-back converter and an L filter. The outer loops are designed by differential flatness: the rotor speed is the flat output of the maximum power point tracking (MPPT) loop, and the energy stored in the DC link and in the filter is the flat output of the DC-link loop. The inner current loops of both converters use passivity-based control (PBC), which keeps the workless coupling of the machine and filter models and injects damping; a storage function proves their stability. An adaptive 6-10-2 neural network, updated online with dead-zone, σ -modification and back-propagation, adds a compensation signal to the flat loops without changing the nominal gains. Simulations under turbulent wind, parameter mismatch, a 10 % error on the aerodynamic model and a 20 % grid voltage dip show accurate MPPT, a DC-link voltage held within ± 0.4 V in nominal conditions, and a 39 % reduction of the integral absolute speed error with the neural compensation.

Index Terms—PMSG, Wind Energy Conversion Systems, Back-to-Back Converter, Flatness-Based Control, Passivity-Based Control, Adaptive Neural Network, Maximum Power Point Tracking (MPPT).

I. INTRODUCTION

Wind energy is one of the main sources in the transition towards low-carbon electricity generation [1]; a record 117 GW of new wind capacity was installed worldwide in 2023 [2]. Direct-drive PMSG turbines remove the gearbox, which reduces maintenance and losses, and their full-scale back-to-back converter decouples the generator from the grid [3], [4]. This converter must extract the maximum power from the wind, regulate the DC-link voltage and inject a clean current into the grid.

These tasks are usually performed by cascaded PI loops in rotating frames [5], [6]. The PMSG model is nonlinear and coupled, and its parameters vary with temperature and saturation. Nonlinear controllers have therefore been studied for wind turbines and PMSGs, such as backstepping [7], predictive control [8] and passivity-based control [9].

Differential flatness [10] computes the control input directly from a desired trajectory of a flat output; using the stored energy as the flat output gives a simple linear power balance for DC buses [11]. Passivity-based control [12], [13] shapes

the energy of the closed loop and keeps the physical structure of the system. Both methods rely on a model, so model errors leave residual tracking errors. Adaptive neural networks can learn such residual terms online [14]; in [15], an adaptive neural compensation was added to a nominal controller of a DFIG without changing its gains.

This paper combines these three tools for the complete chain of a 2 MW direct-drive PMSG turbine: flatness for the speed (MPPT) and DC-link loops, PBC for the current loops of both converters, and the adaptive neural compensation of [15]. The full nonlinear model, with machine data from [16], is given in Sections II–III; the wind profile and the references in Section IV; the control and its stability in Section V; and the simulations in Section VI.

II. WIND TURBINE MODELING

The aerodynamic power captured by the turbine is [17]

$$P_w = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3 \quad (1)$$

where ρ is the air density, R the rotor radius, v the wind speed, β the pitch angle and C_p the power coefficient, which depends on the tip-speed ratio

$$\lambda = \frac{\Omega R}{v} \quad (2)$$

with Ω the rotor speed. The power coefficient is modeled by

$$C_p = 0.5176 \left(\frac{116}{\lambda_i} - 0.4\beta - 5 \right) e^{-21/\lambda_i} + 0.0068\lambda \quad (3)$$

with $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3 + 1)$. For $\beta = 0$, its maximum $C_{p,\max} = 0.48$ is reached at $\lambda_{opt} = 8.1$. The aerodynamic torque is $T_w = P_w/\Omega$, and the direct-drive shaft is described by the one-mass model

$$J \frac{d\Omega}{dt} = T_w - T_e - B \Omega \quad (4)$$

where J is the total inertia, T_e the electromagnetic torque and B the viscous friction coefficient. With $R = 40$ m and $\Omega_n = 2.23$ rad/s, the rated wind speed is $v_n = \Omega_n R/\lambda_{opt} \approx 11$ m/s and the captured power at v_n is about 2 MW.

III. PMSG AND BACK-TO-BACK CONVERTER MODEL

A. PMSG

In the rotor dq frame, with the d axis aligned with the magnet flux, generator convention and $L_d = L_q = L_s$, the PMSG is described by [4]

$$L_s \frac{di_{sd}}{dt} = -R_s i_{sd} + \omega_e L_s i_{sq} - v_{sd} \quad (5)$$

$$L_s \frac{di_{sq}}{dt} = -R_s i_{sq} - \omega_e L_s i_{sd} + \omega_e \psi_f - v_{sq} \quad (6)$$

where $\omega_e = p\Omega$, p is the number of pole pairs and ψ_f the magnet flux. The electromagnetic torque is

$$T_e = \frac{3}{2} p \psi_f i_{sq}. \quad (7)$$

With $\mathbf{i}_s = [i_{sd} \ i_{sq}]^T$, $\mathbf{v}_s = [v_{sd} \ v_{sq}]^T$, $\mathbf{e}_q = [0 \ 1]^T$ and the skew-symmetric matrix $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, the model reads

$$L_s \dot{\mathbf{i}}_s = -R_s \mathbf{i}_s + \omega_e L_s \mathbf{J} \mathbf{i}_s + \omega_e \psi_f \mathbf{e}_q - \mathbf{v}_s. \quad (8)$$

Since $\mathbf{i}_s^T \mathbf{J} \mathbf{i}_s = 0$, the coupling term $\omega_e L_s \mathbf{J} \mathbf{i}_s$ is workless: it exchanges energy between the two axes without producing or dissipating it.

B. Back-to-Back Converter, DC Link and Grid

The two-level converters are represented by their average model. The voltages they synthesize are limited by the space-vector modulation range, $\|\mathbf{v}_s\|, \|\mathbf{v}_i\| \leq V_{dc}/\sqrt{3}$, and, neglecting losses, the DC-link capacitor obeys

$$C V_{dc} \frac{dV_{dc}}{dt} = P_{msc} - P_{gsc} \quad (9)$$

with $P_{msc} = \frac{3}{2} \mathbf{v}_s^T \mathbf{i}_s$ and $P_{gsc} = \frac{3}{2} \mathbf{v}_i^T \mathbf{i}_g$. In the grid-voltage-oriented frame ($v_{gd} = V_m$, $v_{gq} = 0$), the L filter gives

$$L_f \dot{\mathbf{i}}_g = -R_f \mathbf{i}_g + \omega_g L_f \mathbf{J} \mathbf{i}_g + \mathbf{v}_i - \mathbf{v}_g \quad (10)$$

and the powers injected into the grid are $P = \frac{3}{2} v_{gd} i_{gd}$ and $Q = -\frac{3}{2} v_{gd} i_{gq}$.

C. Complete Model

With the state $\mathbf{x} = [i_{sd} \ i_{sq} \ \Omega \ V_{dc} \ i_{gd} \ i_{gq}]^T$, the inputs $\mathbf{u} = [\mathbf{v}_s^T \ \mathbf{v}_i^T]^T$ and the disturbances v and $\mathbf{v}_g$, (4), (8), (9) and (10) form the sixth-order nonlinear model

$$\begin{cases} \dot{i}_{sd} = \frac{1}{L_s} (-R_s i_{sd} + p\Omega L_s i_{sq} - v_{sd}) \\ \dot{i}_{sq} = \frac{1}{L_s} (-R_s i_{sq} - p\Omega L_s i_{sd} + p\Omega \psi_f - v_{sq}) \\ \dot{\Omega} = \frac{1}{J} (T_w(\Omega, v) - \frac{3}{2} p \psi_f i_{sq} - B\Omega) \\ \dot{V}_{dc} = \frac{1}{C V_{dc}} (P_{msc} - P_{gsc}) \\ \dot{i}_{gd} = \frac{1}{L_f} (-R_f i_{gd} + \omega_g L_f i_{gq} + v_{id} - v_{gd}) \\ \dot{i}_{gq} = \frac{1}{L_f} (-R_f i_{gq} - \omega_g L_f i_{gd} + v_{iq} - v_{gq}) \end{cases} \quad (11)$$

The parameters are listed in Table I. The turbine data correspond to a standard 2 MW turbine; the generator, DC-link, filter and grid data are those of the non-salient PMSG of [16]. The magnet flux is consistent with the grid: at 9.75 Hz, the rated frequency of [16], the back-EMF $\psi_f \omega_e = 562$ V equals the peak phase voltage of the 690 V grid.

TABLE I
PARAMETERS OF THE STUDIED SYSTEM

Parameter	Value	Parameter	Value
P_n	2 MW	p	26
R	40 m	R_s	0.78 m Ω
Ω_n	2.23 rad/s	L_s	1.57 mH
T_n	0.897 MN·m	ψ_f	9.18 Wb
$\lambda_{opt}, C_{p,max}$	8.1, 0.48	C	20 mF
ρ	1.225 kg/m ³	V_{dc}^*	1200 V
J	2.8×10^6 kg·m ²	L_f, R_f	0.15 mH, 2 m Ω
B	10^3 N·m·s/rad	Grid	690 V, 50 Hz

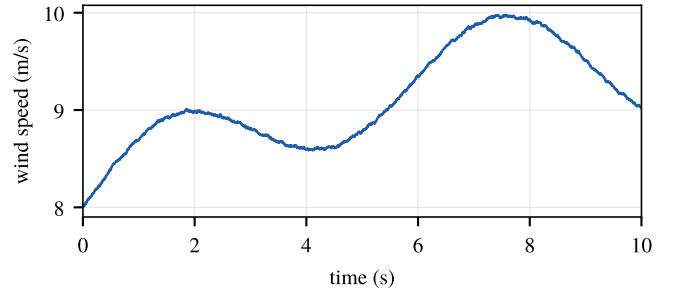


Fig. 1. Wind-speed profile (12).

IV. WIND PROFILE AND REFERENCE GENERATION

The wind speed combines a mean value, low-frequency harmonics and a turbulence term [15]:

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_{turb}(t) \quad (12)$$

with $V_{mean} = 8$ m/s, $A = [1.5, 0.2, 0.3]$ m/s, $f = [0.03, 0.16, 0.17]$ Hz, and v_{turb} a smoothed Gaussian noise (standard deviation 0.01 m/s, time constant 0.05 s). The profile is shown in Fig. 1.

The maximum power is extracted at $\lambda = \lambda_{opt}$. The measured wind is filtered, $\tau_v \dot{\hat{v}} = v - \hat{v}$ with $\tau_v = 0.3$ s, and the speed reference is obtained from the critically damped filter

$$\ddot{\Omega}^* + 2\omega_r \dot{\Omega}^* + \omega_r^2 \Omega^* = \omega_r^2 \frac{\lambda_{opt} \hat{v}}{R} \quad (13)$$

with $\omega_r = 1.5$ rad/s, which provides Ω^* and $\dot{\Omega}^*$ without numerical differentiation. Since T_e only depends on i_{sq} , $i_{sd} = 0$; unity power factor gives $i_{gq}^* = 0$; the DC-link reference is $V_{dc}^* = 1200$ V.

V. PROPOSED CONTROL

The control structure is shown in Fig. 2. It is sampled at $T_s = 100$ μ s.

A. Flatness-Based Outer Loops

A system is flat if its states and inputs can be written as functions of a flat output and of a finite number of its derivatives [10]. The control is then obtained by inverting the model with a desired dynamics imposed on the flat output.

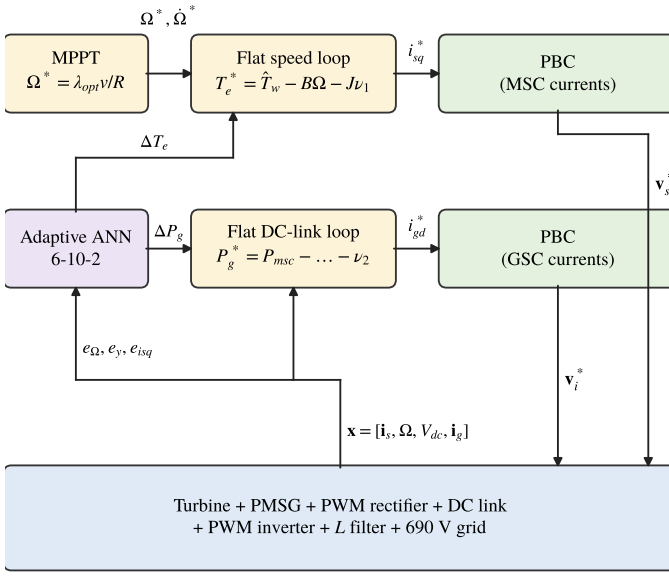


Fig. 2. Proposed control: flat outer loops, PBC current loops and adaptive neural compensation.

Speed loop. From (4), $T_e = T_w - B\Omega - J\dot{\Omega}$, so $y_1 = \Omega$ is a flat output. Replacing $\dot{\Omega}$ by

$$\nu_1 = \dot{\Omega}^* + k_{11}e_\Omega + k_{12} \int e_\Omega dt, \quad e_\Omega = \Omega^* - \Omega \quad (14)$$

gives

$$T_e^* = \hat{T}_w - B\Omega - J\nu_1 + \Delta T_e, \quad i_{sq}^* = \frac{T_e^*}{\frac{3}{2}p\psi_f} \quad (15)$$

where $\hat{T}_w$ is computed from (1)–(3) with the measured wind and ΔT_e is the neural compensation. T_e^* is limited to $[0, 1.2T_n]$ and the integral is frozen during saturation. If $\hat{T}_w = T_w$ and $\Delta T_e = 0$, the error obeys $\dot{e}_\Omega + k_{11}e_\Omega + k_{12} \int e_\Omega dt = 0$.

DC-link loop. The flat output is the energy stored in the DC link and in the filter,

$$y_2 = \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2. \quad (16)$$

From (9) and (10), using $\mathbf{i}_g^T \mathbf{J} \mathbf{i}_g = 0$,

$$\dot{y}_2 = P_{msc} - \frac{3}{2}v_{gd}i_{gd} - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 \quad (17)$$

which is linear in the injected power, as in [11]. Imposing $\dot{y}_2 = -\nu_2$ with $\nu_2 = k_{21}e_y + k_{22} \int e_y dt$, $e_y = y_2^* - y_2$, yields

$$P_g^* = P_{msc} - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 - \nu_2 + \Delta P_g, \quad i_{gd}^* = \frac{2P_g^*}{3v_{gd}}. \quad (18)$$

The term P_{msc} , known from the applied voltages and measured currents, anticipates the power coming from the generator. The gains are $k_{11} = 2\zeta_\Omega\omega_\Omega$, $k_{12} = \omega_\Omega^2$, $k_{21} = 2\zeta_v\omega_v$, $k_{22} = \omega_v^2$ (Table II).

B. Passivity-Based Current Loops

For the machine-side converter, the error $\mathbf{e}_s = \mathbf{i}_s - \mathbf{i}_s^*$ and its integral $\mathbf{z}_s (\dot{\mathbf{z}}_s = \mathbf{e}_s)$ are used in

$$\dot{\mathbf{v}}_s^* = -L_s\dot{\mathbf{i}}_s^* - R_s\mathbf{i}_s^* + \omega_e L_s \mathbf{J} \mathbf{i}_s^* + \omega_e \psi_f \mathbf{e}_q + R_a \mathbf{e}_s + K_i \mathbf{z}_s \quad (19)$$

where R_a is the injected damping. For the grid-side converter, with $\mathbf{e}_g = \mathbf{i}_g - \mathbf{i}_g^*$ and $\dot{\mathbf{z}}_g = \mathbf{e}_g$,

$$\dot{\mathbf{v}}_i^* = L_f \dot{\mathbf{i}}_g^* + R_f \mathbf{i}_g^* - \omega_g L_f \mathbf{J} \mathbf{i}_g^* + \mathbf{v}_g - R_b \mathbf{e}_g - K_{ig} \mathbf{z}_g. \quad (20)$$

The reference derivatives are obtained by a first-order filter ($\tau_d = 2$ ms) and the voltages are limited to $V_{dc}/\sqrt{3}$, with the integrals frozen at the limit.

Proposition 1. With (19) applied to (8) and $R_a, K_i > 0$, the storage function $H = \frac{1}{2}L_s\|\mathbf{e}_s\|^2 + \frac{1}{2}K_i\|\mathbf{z}_s\|^2$ satisfies $\dot{H} = -(R_s + R_a)\|\mathbf{e}_s\|^2 \leq 0$, and $\mathbf{e}_s \rightarrow 0$. The same holds for (20) applied to (10).

Proof. Subtracting $L_s\dot{\mathbf{i}}_s^*$ from (8) and inserting (19) gives $L_s\dot{\mathbf{e}}_s = -(R_s + R_a)\mathbf{e}_s + \omega_e L_s \mathbf{J} \mathbf{e}_s - K_i \mathbf{z}_s$. Hence $\dot{H} = \mathbf{e}_s^T L_s \dot{\mathbf{e}}_s + K_i \mathbf{z}_s^T \mathbf{e}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2 + \omega_e L_s \mathbf{e}_s^T \mathbf{J} \mathbf{e}_s$, and the last term vanishes because $\mathbf{J}$ is skew-symmetric. H is bounded and $\dot{H}$ is uniformly continuous, so Barbalat's lemma [18] gives $\mathbf{e}_s \rightarrow 0$. $\square$

C. Adaptive Neural Compensation

As in [15], the network does not replace the nominal control u_N given by (15)–(18) and does not modify its gains; it adds

$$u = u_N + u_{AI}, \quad u_{AI} = [\Delta T_e \ \Delta P_g]^T. \quad (21)$$

Its six inputs are the normalized errors and their filtered derivatives, $\mathbf{z}_n = S_{in}^{-1}[e_\Omega, e_y, e_{isq}, \dot{e}_\Omega, \dot{e}_y, \dot{e}_{isq}]^T$. With one hidden layer of ten neurons,

$$\mathbf{h} = \tanh(W_1 \mathbf{z}_n + \mathbf{b}_1); \mathbf{y} = W_2 \mathbf{h} + \mathbf{b}_2; u_{AI} = K_{AI} \text{sat}(\mathbf{y}) \quad (22)$$

with $W_1 \in \mathbb{R}^{10 \times 6}$, $W_2 \in \mathbb{R}^{2 \times 10}$, scaled by $0.1T_n$ and $0.1P_n$. T_e and P_g enter $\dot{e}_\Omega$ and $\dot{e}_y$ with a positive sign, so the learning signal in the control direction is $\mathbf{r} = B^T \mathbf{e} = [e_\Omega \ e_y]^T$ (normalized), set to zero inside a dead zone $\|\mathbf{r}\| \leq \delta$. The parameters are updated online by

$$\dot{W}_2 = -\Gamma \mathbf{r} \mathbf{h}^T - \sigma W_2, \quad \dot{\mathbf{b}}_2 = -\Gamma \mathbf{r} - \sigma \mathbf{b}_2, \quad (23)$$

$$\dot{W}_1 = -\Gamma \delta_h \mathbf{z}_n^T - \sigma W_1, \quad \dot{\mathbf{b}}_1 = -\Gamma \delta_h - \sigma \mathbf{b}_1, \quad (24)$$

with $\delta_h = (1 - \mathbf{h}^2) \odot (W_2^T \mathbf{r})$. The σ -modification terms [14], [19] prevent parameter drift. The adaptation is frozen when T_e^* or i_{gd}^* saturates, and u_{AI} is applied at the next sample. W_2 and $\mathbf{b}_2$ start at zero, so the compensation is initially inactive.

Theorem 1 ([15]). Assume that the nominal control makes the tracking error exponentially stable, that the residual to be compensated is approximated by the network with a bounded error on the operating region, and that $\Gamma, \sigma > 0$. Then, under the above laws with the dead zone, the tracking error is uniformly ultimately bounded.

Remark 1. In simulation, $\Gamma \geq 20$ destabilizes the DC-link loop; the bound of the theorem requires bounded network

TABLE II
CONTROLLER AND ADAPTIVE ANN PARAMETERS

Parameter	Value
Speed loop	$\omega_\Omega = 2 \text{ rad/s}$, $\zeta_\Omega = 0.9$ ($k_{11} = 3.6$, $k_{12} = 4$)
DC-link loop	$\omega_v = 60 \text{ rad/s}$, $\zeta_v = 1$ ($k_{21} = 120$, $k_{22} = 3600$)
PBC (MSC)	$R_a = 0.785 \text{ } \Omega$, $K_i = 98.1 \text{ } \Omega/\text{s}$
PBC (GSC)	$R_b = 0.075 \text{ } \Omega$, $K_{ig} = 9.38 \text{ } \Omega/\text{s}$
ANN architecture	6–10–2
Γ , σ , δ , K_{AI}	5, 0.01, 0.1, 1
Output saturation	± 3 ($\pm 0.3T_n$, $\pm 0.3P_n$)

TABLE III
TRACKING PERFORMANCE (10 s)

Metric	Nominal	Robustness	
		without ANN	with ANN
IAE of Ω (rad)	2.7×10^{-6}	0.0109	0.0067
RMS error of Ω (rad/s)	3.3×10^{-7}	2.1×10^{-3}	1.2×10^{-3}
RMS error of i_{sq} (A)	0.80	3.05	3.08
RMS error of i_{gd} (A)	2.75	6.88	7.61
ISE of V_{dc} (V^2s)	0.085	1.30	1.16
Max $ V_{dc} - V_{dc}^* $, $t < 1 \text{ s}$ (V)	0.31	10.44	10.42
Max $ V_{dc} - V_{dc}^* $, dip (V)	–	3.35	3.60
Mean C_p	0.4763	0.4764	0.4763

parameters, which a moderate adaptation gain ensures. $\Gamma = 5$ is used.

VI. NUMERICAL SIMULATIONS

The model (11) is integrated with a fourth-order Runge–Kutta scheme at step T_s over 10 s, starting from the MPPT operating point at $v = 8 \text{ m/s}$. Two cases are considered:

- *nominal*: the controller uses the true parameters;
- *robustness*: the controller keeps the nominal values while the plant has $R_s \times 1.5$, $L_s \times 1.2$, $\psi_f \times 0.92$, $R_f \times 1.5$, $L_f \times 1.25$; $\hat{T}_w$ is 10 % lower than T_w ; and the grid voltage drops by 20 % between 8 and 8.2 s.

The robustness case is run with and without the neural compensation, with the same nominal gains, so that the difference is due to the network only. The results are summarized in Table III.

Nominal case. The speed follows the MPPT reference with an RMS error of $3.3 \times 10^{-7} \text{ rad/s}$, C_p stays between 0.47 and 0.48, and the DC-link voltage remains within $\pm 0.36 \text{ V}$ of 1200 V, owing to the anticipation of P_{msc} in (18). The errors remain inside the dead zone, so the network does not adapt and both versions give the same results.

Robustness case. Figs. 3 and 4 show that the speed and the currents follow their references, that Q stays near zero and that C_p remains between 0.47 and 0.48 (mean 0.476). The largest DC-link deviation, 10.4 V, occurs during the first instants, when the controller starts with mismatched parameters. During the 20 % grid voltage dip, the deviation is 3.6 V and the voltage returns within 1 V of its reference less than 40 ms after each voltage step.

Fig. 5 shows that, without knowing the 10 % model error, the network converges in less than one second to $\Delta T_e \approx$

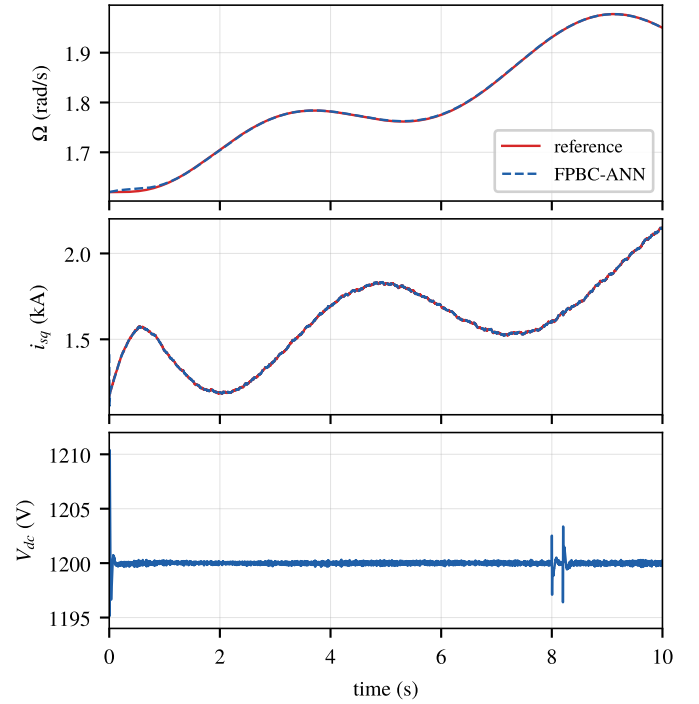


Fig. 3. Robustness case with the proposed control: rotor speed, q -axis generator current and DC-link voltage.

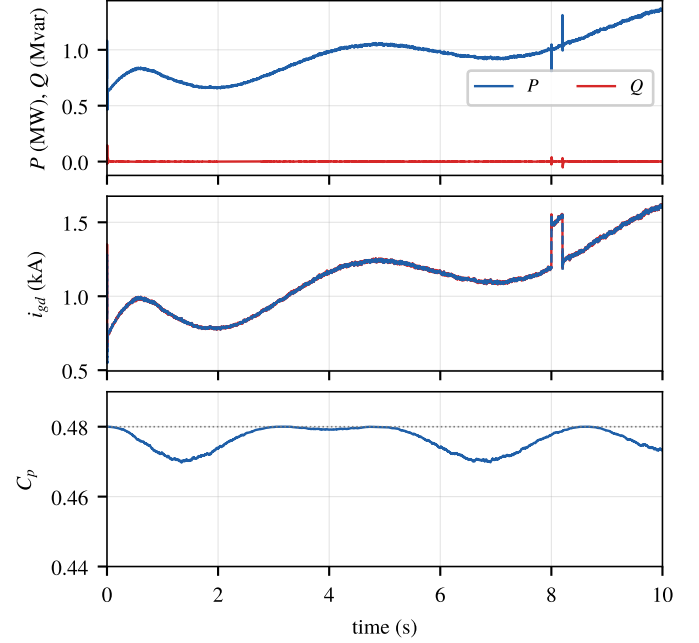


Fig. 4. Robustness case: powers injected into the grid, d -axis grid current and power coefficient.

0.065 MN·m, close to the mean of the actual error $T_w - \hat{T}_w$ (between 0.047 and 0.077 MN·m); the slowly varying part is handled by the integral action of (15). With the same nominal gains, the network reduces the IAE of the speed by 39 % and the ISE of the DC-link voltage by 11 %, mainly during the

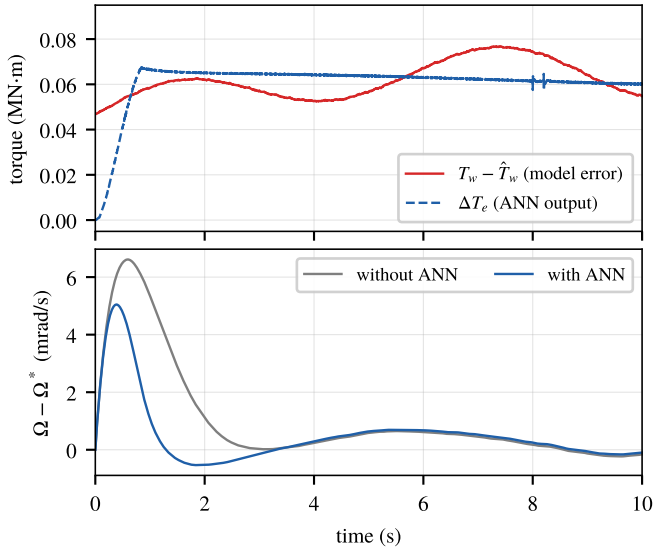


Fig. 5. Robustness case: neural compensation ΔT_e compared with the aerodynamic model error, and speed error with and without the network.

start-up transient, where the peak speed error drops from 6.6 to 5.1 mrad/s. The current errors and the dip deviation are slightly larger with the network (Table III), and the mean C_p is unchanged: the benefit is in tracking accuracy, not in extracted energy.

A switched model of the two-level converters (2160 Hz carrier) was also simulated at 10 m/s: the THD of the grid current is 2.3 %, below the 5 % limit of IEEE 519 [20].

VII. CONCLUSION

A complete model of a 2 MW direct-drive PMSG wind energy conversion system, from the turbine to the grid, was developed with machine data from the literature. Its control combines flatness for the speed and DC-link energy loops, passivity-based control for the current loops of both converters, whose stability is proven with a storage function, and an adaptive neural compensation that does not change the nominal gains. Simulations show accurate MPPT under turbulent wind and a tightly regulated DC link; under parameter and aerodynamic model errors, the network learns the torque error online and reduces the speed tracking error by 39 %. Future work will address pitch control above rated wind, non-ideal synchronization and experimental validation.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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